# Class 10: Structural Bioinformatics (pt. 1)

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## Table of contents

The PDB database . . . . . 1
Using Molstar . . . . . 2
The Bio3D package for structural bioinformatics . . . . . 5
Quick viewing of PDBs . . . . . 6
Prediction of Protein flexibility . . . . . 7
Comparative protein structure analysis with PCA . . . . . 8

### The PDB database

The Protein Data Bank (PDB) is the main repository of biomolecular structure data. Lets see what is in it:

```
stats <- read.csv("pdbstats26", row.names = 1)
stats
```

```
                         X.ray    EM   NMR Integrative Multiple.methods Neutron
Protein (only)          178795 21825 12773         343               226      84
Protein/Oligosaccharide  10363  3564    34           8                11       1
Protein/NA                9106  6335   287          24                 7       0
Nucleic acid (only)       3132   221  1566           3                15       3
Other                      175    25    33           4                 0       0
Oligosaccharide (only)      11     0     6           0                 1       0
                        Other  Total
Protein (only)             32 214078
Protein/Oligosaccharide     0  13981
Protein/NA                  0  15759
Nucleic acid (only)         1   4941
Other                       0    237
Oligosaccharide (only)      4     22
```

```r
n.sums <- colSums(stats)
n <- n.sums/n.sums["Total"]
round(n, digits=2)
```

```
           X.ray               EM              NMR      Integrative
            0.81             0.13             0.06             0.00
Multiple.methods          Neutron            Other            Total
            0.00             0.00             0.00             1.00
```

> Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

X-Ray: 81% and Electron Microscopy: 13%

> Q2: What proportion of structures in the PDB are protein?

```r
protein_total <- sum(stats[c("Protein (only)", "Protein/NA"), "Total"])
protein_prop <- protein_total / sum(stats[, "Total"])

round(protein_prop, 2)
```

```
[1] 0.92
```

> Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB? What is the total number of entries in the PDB

```r
n.sums["Total"]
```

```
 Total
249018
```

### Using Molstar

We can use the main Molstar viewer online:

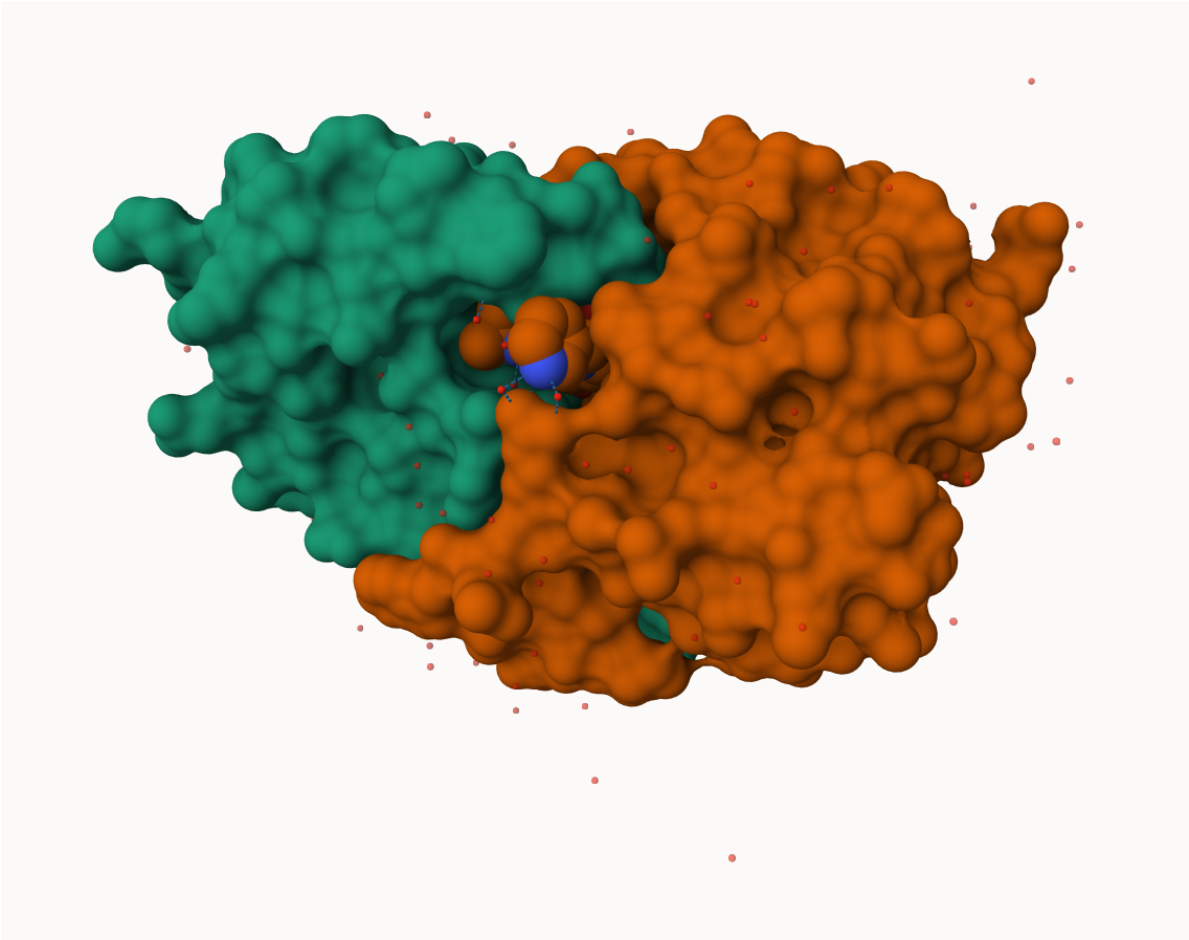
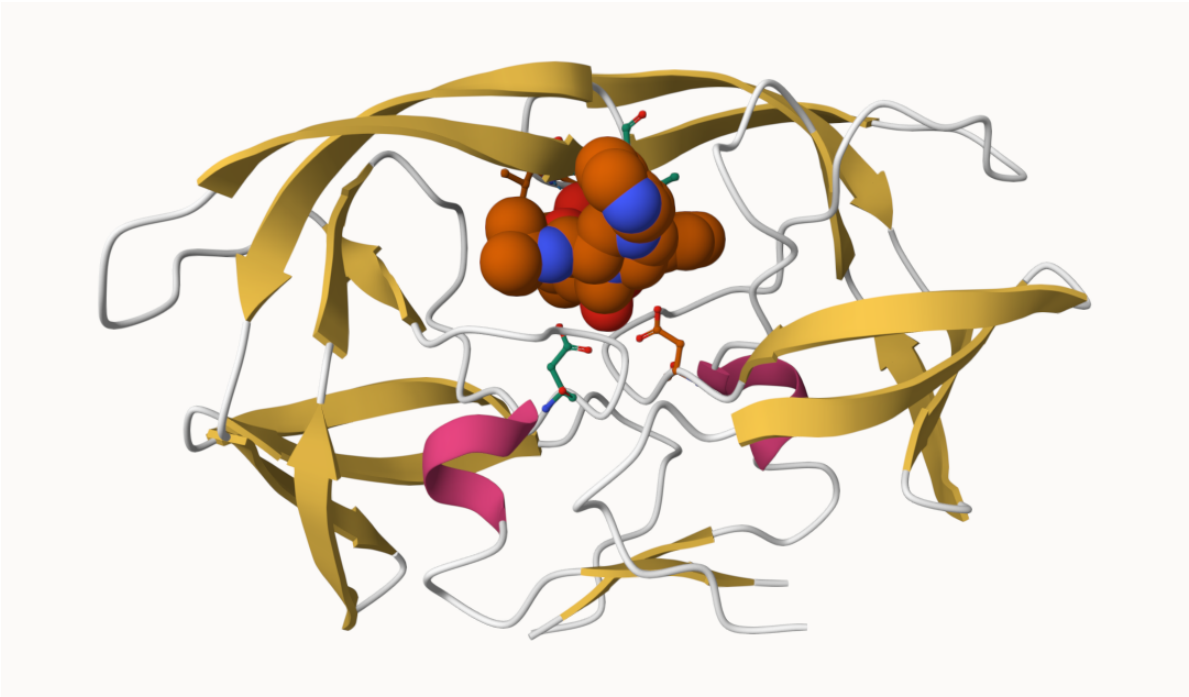
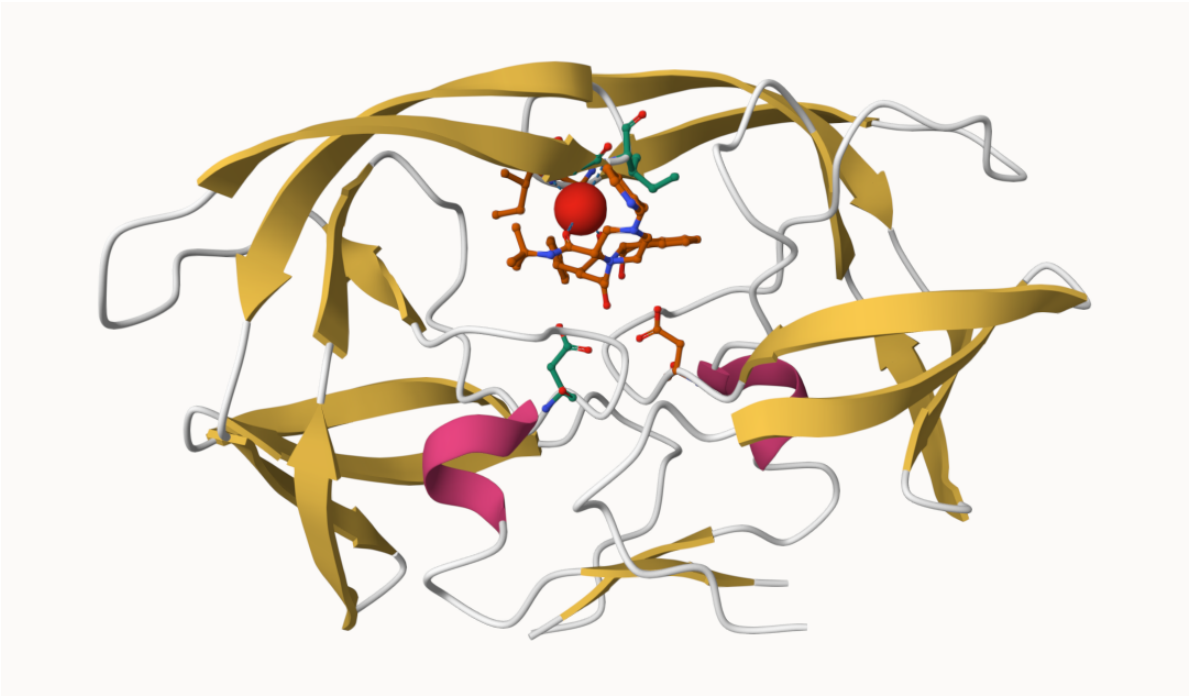


Figure 1: First view of HIV-Pr dimer with bound inhibitor

Q. Generate and insert an image of the HIV-Pr cartoon colored by secondary structure, showing inhibitor (ligand) in ball and stick.



Q. One final image showing catalytic APS 25 and the all-important active site water molecule as spacefill.



### The Bio3D package for structural bioinformatics

```r
library(bio3d)

hiv <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
hiv
```

```
 Call:  read.pdb(file = "1hsg")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
head( hiv$atom )
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
```

```
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

```
pdbseq(hiv)
```

```
  1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20
"P" "Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K"
 21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40
"E" "A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G"
 41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60
"R" "W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D"
 61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80
"Q" "I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T"
 81  82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99   1
"P" "V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F" "P"
  2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20  21
"Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K" "E"
 22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40  41
"A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G" "R"
 42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60  61
"W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D" "Q"
 62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80  81
"I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T" "P"
 82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99
"V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F"
```

Let's try out the new **bio3dview** package that is not yet in CRAN. We can use the **remotes** package to install any R packages from GitHub.

### Quick viewing of PDBs

```
library(bio3dview)

sele <- atom.select(hiv, resno=25)
```

```
#view.pdb(hiv, backgroundColor = "pink",
#         highlight = sele,
#         highlight.style = "spacefill")
```

**Prediction of Protein flexibility**

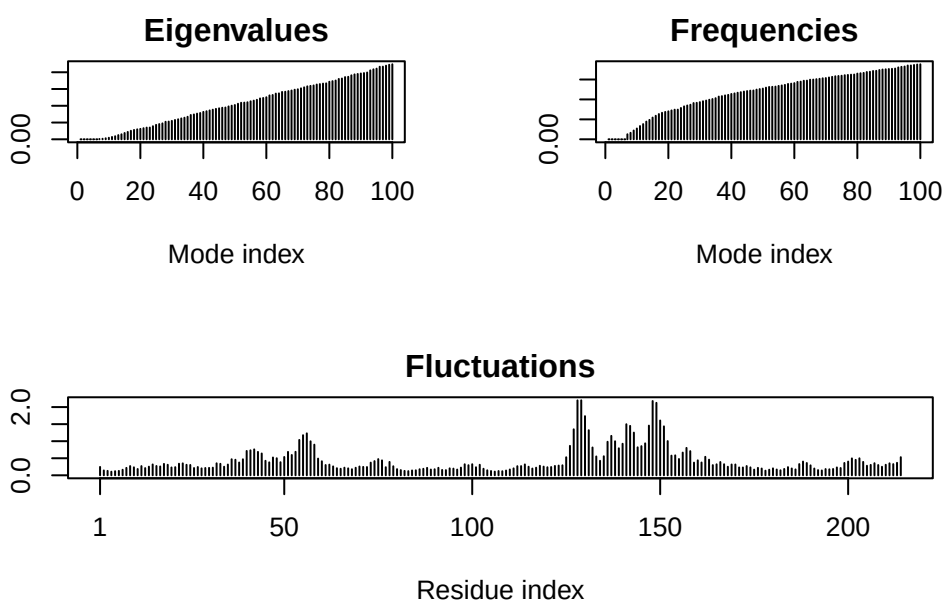
```
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

```
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.013 seconds.
 Diagonalizing Hessian...   Done in 0.047 seconds.
```

```
plot(m)
```




Write out our results as a wee trajectory movie:

```r
mktrj(m, file="results.pdb")
```

```r
#view.nma(m)
```

### Comparative protein structure analysis with PCA

We will start with a database id "1ake_A"

```r
library(bio3d)

id <- "1ake_A"
aa <- get.seq(id)
```

```
Warning in get.seq(id): Removing existing file: seqs.fasta


Fetching... Please wait. Done.
```

```r
aa
```

```
             1        .         .         .         .         .         60
pdb|1AKE|A   MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
             1        .         .         .         .         .         60

            61        .         .         .         .         .        120
pdb|1AKE|A   DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
            61        .         .         .         .         .        120

           121        .         .         .         .         .        180
pdb|1AKE|A   VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
           121        .         .         .         .         .        180

           181        .         .         .   214
pdb|1AKE|A   YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
           181        .         .         .   214

Call:
  read.fasta(file = outfile)

Class:
```

```
  fasta

Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call
```

```r
blast <- blast.pdb(aa)
```

```
 Searching ... please wait (updates every 5 seconds) RID = STJV183G014
 ......................................................................
 Reporting 96 hits
```

Have a wee peak:

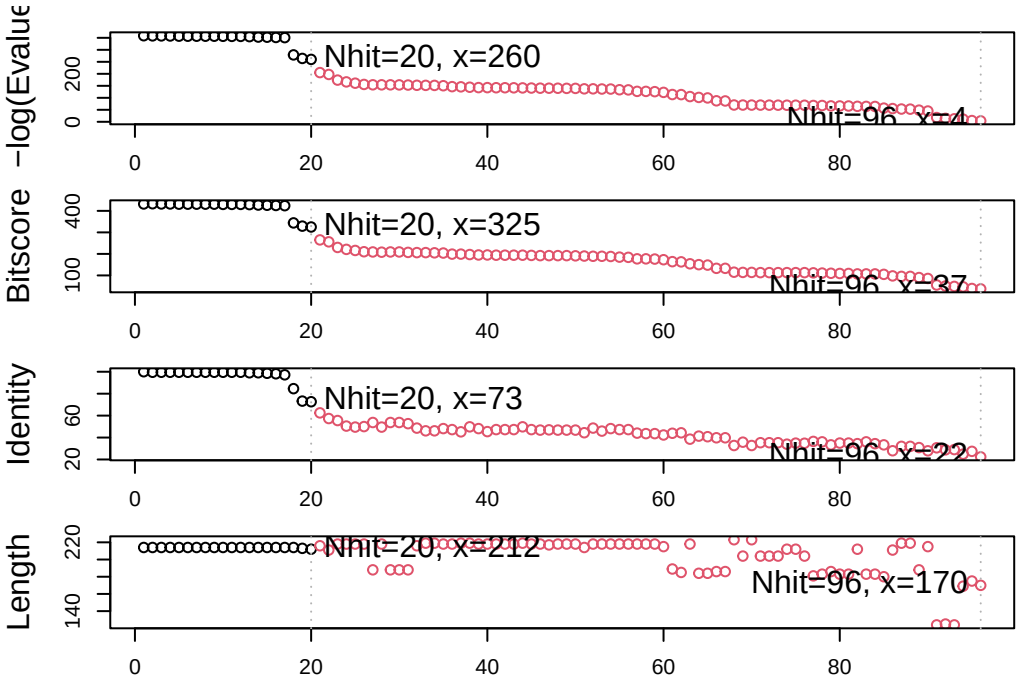
```r
head( blast$hit.tbl )
```

```
        queryid subjectids identity alignmentlength mismatches gapopens q.start
1 Query_2286183     1AKE_A  100.000             214          0        0       1
2 Query_2286183     8BQF_A   99.533             214          1        0       1
3 Query_2286183     4X8M_A   99.533             214          1        0       1
4 Query_2286183     6S36_A   99.533             214          1        0       1
5 Query_2286183     9R6U_A   99.533             214          1        0       1
6 Query_2286183     9R71_A   99.533             214          1        0       1
  q.end s.start s.end    evalue bitscore positives mlog.evalue pdb.id    acc
1   214       1   214 1.78e-156      432    100.00    358.6267 1AKE_A 1AKE_A
2   214      21   234 2.91e-156      433    100.00    358.1351 8BQF_A 8BQF_A
3   214       1   214 3.18e-156      432    100.00    358.0464 4X8M_A 4X8M_A
4   214       1   214 4.67e-156      432    100.00    357.6621 6S36_A 6S36_A
5   214       1   214 1.04e-155      431     99.53    356.8615 9R6U_A 9R6U_A
6   214       1   214 1.23e-155      431     99.53    356.6937 9R71_A 9R71_A
```

```r
hits <- plot(blast)
```

```
  * Possible cutoff values:    260 3
            Yielding Nhits:    20 96

  * Chosen cutoff value of:    260
            Yielding Nhits:    20
```




Peak at our "top hits"

```
head ( hits$pdb.id )
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A"
```

Now we can download these "top hits" these will all be ADK structures in the ADK database.